Weekly Progress Report

Week 03 — Data Science & Machine Learning Internship

STUDENT NAME

Udit Kumar

DOMAIN

Data Science and Machine Learning

SELECTED PROJECTS

- **Agriculture:** Prediction of Agriculture Crop Production in India
- **Smart City:** Forecasting of Smart City Traffic Patterns

1. Tasks & Progress

- **Milestones Achieved:**
  - Studied and integrated statistical theory from *An Introduction to Probability and Statistics* into the exploratory and inferential phases of both internship projects.
  - Formulated hypothesis testing frameworks and applied probability distribution modeling to validate dataset assumptions.
- **Key Work Completed:**
  - **Agriculture Project:** Evaluated crop yield variables using Parametric Estimation (Method of Moments & MLE) and Nonparametric Statistical Inference (Chi-square test of independence and Spearman's rank correlation) to assess the relationship between regional climate factors and yield outputs.
  - **Smart City Traffic Project:** Applied Hypothesis Testing (t-tests and F-tests) to evaluate variations in peak vs. non-peak hourly traffic flows, and conducted Kolmogorov–Smirnov tests to check traffic distribution normality.

2. Challenges & Hurdles

- **Obstacles Encountered:**
  - Deviations from standard normality assumptions in real-world time-series traffic and crop yield distributions.
  - Complexity in applying parametric tests when dataset variances were non-homogeneous across regional subsets.
- **Solutions Applied:**
  - Shifted to Nonparametric Statistical Inference methods (Mann–Whitney–Wilcoxon test and Kolmogorov–Smirnov test) when parametric assumptions failed.
  - Utilized Central Limit Theorem principles and Large Sample Theory for asymptotic approximations where sample sizes were large enough.

3. Lessons Learned

- **Hypothesis Testing & Inference:** Mastered practical implementations of Neyman–Pearson theory, t-tests, F-tests, and Chi-square independence tests for dataset validation.
- **Parametric vs. Nonparametric Methods:** Learned how and when to select nonparametric alternatives (Mann–Whitney, KS-test, Spearman's rank correlation) over classical parametric procedures.
- **Statistical Foundations in ML:** Gained a deep understanding of probability distributions, point estimation techniques (MLE), and confidence intervals to ensure robust feature selection in predictive modeling.